vingnette

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1 SIMPLE SEDML VINGNETTE

This notebook provides examples of using `SimpleSBML`, a Python system for creating and validating Simulation Experiment Description Markup Language (SED-ML) descriptions of computational experiments for models written in the Systems Biology Markup Language (SBML) community standard.

Throughout, we use the term **display variable** to refer to a subset of variables in the model that the user wants to observe (e.g., floating species). A **parameter scan** variable is a parameter of the model that is being varied to see its effect on display variables.

SimpleSEDML provides APIs at two levels. The high level API is task oriented. Four tasks are considered. * Time course simulation for a single model. In this task, the user wants to see the interaction between display variables over time. * Time course simulation for multiple models. Here, the user wants to compare the time courses of the same display variable in different models. * Parameter scan for a single model. The user wants to see how display values are affected by changes in scan parameters. * Parameter scan for multiple models. The user wants to compare the effect on the same display variable in different models as a scan parameter is changed.

The above use cases are encapsulated into 4 different API calls with optional parameters related to the simulation and changing values of the paramters of the model. These API calls produce an object with the following methods: * `getSEDML` produces SED-ML for the experiment. * `getPhraSEDML` produces the PhraSED-ML for the experiment. * `execute` returns a `DataFrame` and a plot of the variables specified in the constructor. * `makeOMEXFile` creates an omex file

The low level API is essentially a python wrapper for PhraSED-ML, and is available from all objects produced in the high level API.

Below, we illustrate each one of the high level APIs.

2 Preliminaries

[1]:
```
# Handle execution locally and in Colab
try:
    import SimpleSEDML as ss
except:
    !pip install SimpleSEDML
    import SimpleSEDML as ss
import pandas as pd
```

```
/Users/jlheller/home/Technical/repos/SimpleSEDML/ss/lib/python3.11/site-
packages/kisao/core.py:11: UserWarning: pkg_resources is deprecated as an API.
See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources
package is slated for removal as early as 2025-11-30. Refrain from using this
package or pin to Setuptools<81.
  import pkg_resources
```

[2]:
```
MODEL_ANT = '''
model myModel
    J1: S1 -> S2; k1*S1
    J2: S2 -> S3; k2*S2
    S1 = 10; S2 = 0
    k1 = 1;k2 = 1
    S1 is "species1"
    S2 is "species2"
end
'''
```

[3]:
```
MODEL2_ANT = '''
model myModel
    J1: S1 -> S2; k1*S1
    J2: S2 -> S3; k2*S2
    J3: S3 -> S1; k3*S3

    S1 = 10
    S2 = 0
    k1 = 1
    k2 = 1
    k3 = 0.1
end
'''
```

3 Use Case 1: Single module time course

3.1 Default parameters

[4]:
```
?ss.makeSingleModelTimeCourse
```

```
Init signature:
ss.makeSingleModelTimeCourse(
    model_ref: str,
    project_id: Optional[str] = None,
    ref_type: Optional[str] = None,
    simulation_type: str = 'uniform',
    project_dir: Optional[str] = None,
    display_variables: Optional[List[str]] = None,
    start: float = 0,
```

```
    end: float = 5,
    num_step: Optional[int] = None,
    num_point: Optional[int] = None,
    title: Optional[str] = None,
    algorithm: Optional[str] = None,
    is_plot: bool = True,
    model_parameter_dct: Optional[dict] = None,
)
Docstring:      Class to create a time course simulation for a single model
Init docstring:
Creates a time course simulation

Args:
    model_ref: reference to the model
    project_id: ID of the project, if None, uses the default project ID
    ref_type: type of the reference (e.g. "sbml_str", "ant_str", "sbml_file",␣
 ↪"ant_file", "sbml_url")
    simulation_type: type of the simulation
            (e.g., "uniform", "uniform_stochastic", "steadystate", "onestep")
    project_dir: directory to save the files
    display_variables: variables to be plotted and included the report
    start: start time
    end: end time
    num_step: number of steps
    num_point: number of points to be plotted
    algorithm: algorithm to use for the simulation
    title: title of the plot
    is_plot: if True, plot the results
    model_parameter_dct: dictionary of parameters whose values are changed

Returns:
    SingleModelTimeCourse: a time course simulation object
File:           ~/home/Technical/repos/SimpleSEDML/src/SimpleSEDML/
 ↪single_model_time_course.py
Type:           type
Subclasses:
```

[5]:
```
smtc = ss.makeSingleModelTimeCourse(MODEL_ANT, title="My Plot", num_point=100)
```

[6]:
```
print(smtc.getPhraSEDML())
```

```
project_model = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmpm1d__nvp/project_model.xml"
project_sim = simulate uniform(0, 5.0, 99)
project_sim.algorithm = CVODE
project_task = run project_sim on project_model
report time, project_task.S1, project_task.S2, project_task.S3
plot "My Plot" time vs project_task.S1, project_task.S2, project_task.S3
```

[7]:
```python
print(smtc.getSEDML())
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Created by phraSED-ML version v1.3.0 with libSBML version 5.19.5. -->
<sedML xmlns="http://sed-ml.org/sed-ml/level1/version4"
xmlns:sbml="http://www.sbml.org/sbml/level3/version2/core" level="1"
version="4">
  <listOfModels>
    <model id="project_model"
language="urn:sedml:language:sbml.level-3.version-2" source="/var/folders/02/lb2
48q1j50dch2pthzkvxmpr0000gn/T/tmpm1d__nvp/project_model.xml"/>
  </listOfModels>
  <listOfSimulations>
    <uniformTimeCourse id="project_sim" initialTime="0" outputStartTime="0"
outputEndTime="5" numberOfSteps="99">
      <algorithm name="CVODE" kisaoID="KISAO:0000019"/>
    </uniformTimeCourse>
  </listOfSimulations>
  <listOfTasks>
    <task id="project_task" modelReference="project_model"
simulationReference="project_sim"/>
  </listOfTasks>
  <listOfDataGenerators>
    <dataGenerator id="report_0_0_0" name="time">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> time </ci>
      </math>
      <listOfVariables>
        <variable id="time" symbol="urn:sedml:symbol:time"
taskReference="project_task" modelReference="project_model"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_1" name="species1">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> project_task_____S1 </ci>
      </math>
      <listOfVariables>
        <variable id="project_task_____S1"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S1']"
taskReference="project_task" modelReference="project_model"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_2" name="species2">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> project_task_____S2 </ci>
      </math>
      <listOfVariables>
```

```
        <variable id="project_task_____S2"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S2']"
taskReference="project_task" modelReference="project_model"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_3" name="S3">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> project_task_____S3 </ci>
      </math>
      <listOfVariables>
        <variable id="project_task_____S3"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S3']"
taskReference="project_task" modelReference="project_model"/>
      </listOfVariables>
    </dataGenerator>
  </listOfDataGenerators>
  <listOfOutputs>
    <report id="report_0">
      <listOfDataSets>
        <dataSet id="report_0_0_0_dataset" label="time"
dataReference="report_0_0_0"/>
        <dataSet id="report_0_0_1_dataset" label="project_task.S1"
dataReference="report_0_0_1"/>
        <dataSet id="report_0_0_2_dataset" label="project_task.S2"
dataReference="report_0_0_2"/>
        <dataSet id="report_0_0_3_dataset" label="project_task.S3"
dataReference="report_0_0_3"/>
      </listOfDataSets>
    </report>
    <plot2D id="plot_1" name="My Plot">
      <listOfCurves>
        <curve id="plot_1__report_0_0_0__report_0_0_1" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_1"/>
        <curve id="plot_1__report_0_0_0__report_0_0_2" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_2"/>
        <curve id="plot_1__report_0_0_0__report_0_0_3" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_3"/>
      </listOfCurves>
    </plot2D>
  </listOfOutputs>
</sedML>
```

We can also look at the PhraSED-ML that gets generated by changing the object that gets returned.

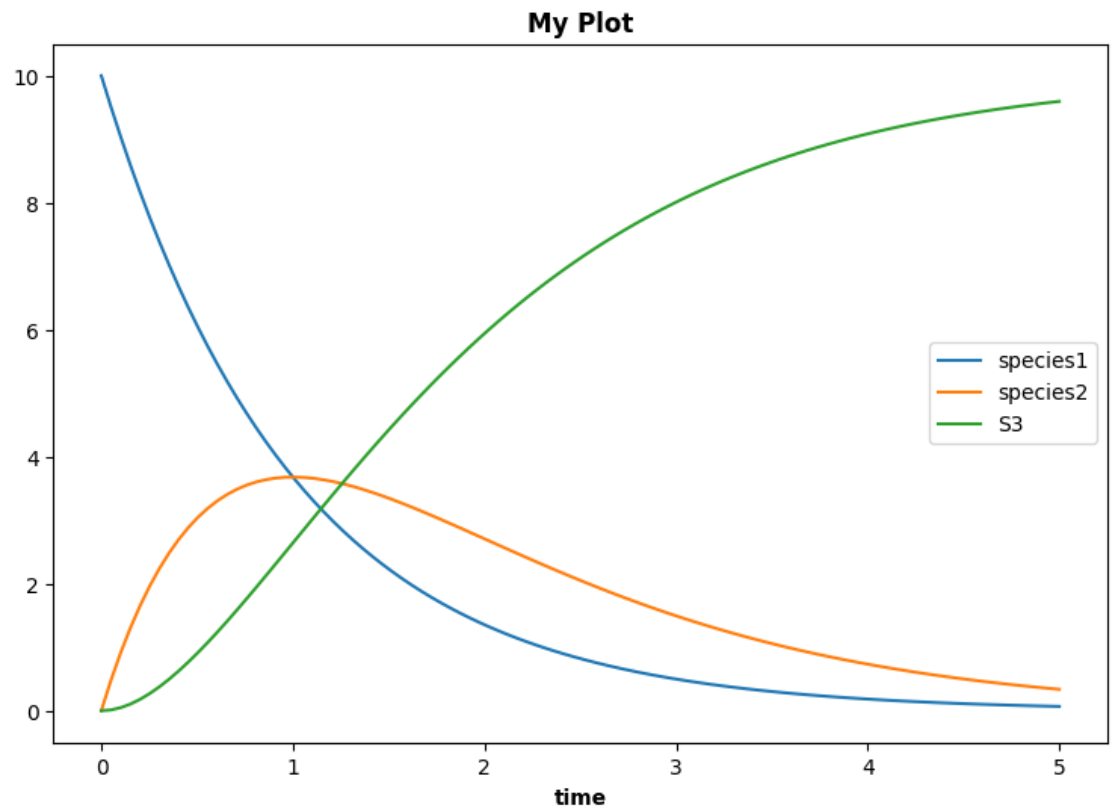
[8]:
```
print(smtc.getPhraSEDML())
```

```
project_model = model
```

```
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmpm1d__nvp/project_model.xml"
project_sim = simulate uniform(0, 5.0, 99)
project_sim.algorithm = CVODE
project_task = run project_sim on project_model
report time, project_task.S1, project_task.S2, project_task.S3
plot "My Plot" time vs project_task.S1, project_task.S2, project_task.S3
```

And we can execute the SEDML, getting both a plot and a DataFrame of time course values.

[9]:
```
df = smtc.execute()
```



[10]:
```
df.head()
```

[10]:

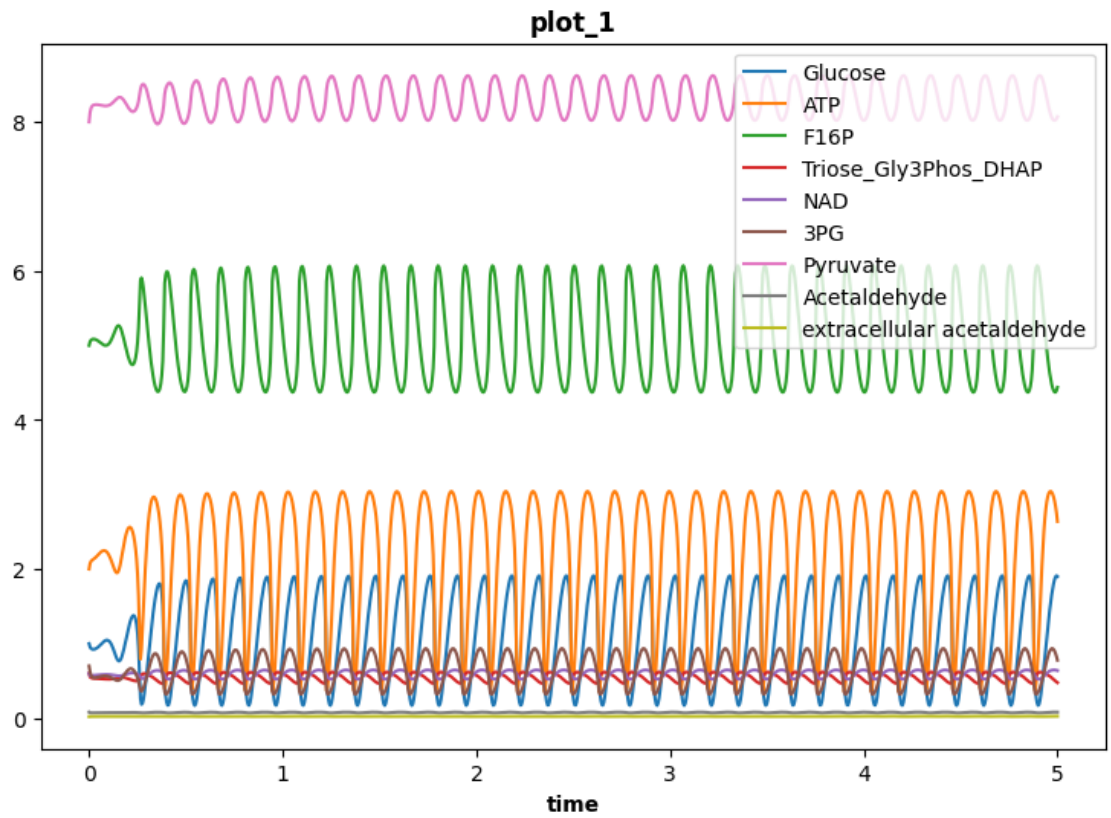
	time	project_task.S1	project_task.S2	project_task.S3
0	0.000000	10.000000	0.000000	0.000000
1	0.050505	9.507491	0.480176	0.012332
2	0.101010	9.039239	0.913054	0.047707
3	0.151515	8.594049	1.302129	0.103823
4	0.202020	8.170784	1.650664	0.178552

[11]:
```
# Eliminate the temporary files
smtc.cleanUp()
```

3.2 Non-default arguments

[12]:
```
WOLF_URL = "https://raw.githubusercontent.com/sys-bio/temp-biomodels/refs/heads/
↪main/final/BIOMD0000000206/BIOMD0000000206_url.xml"
```

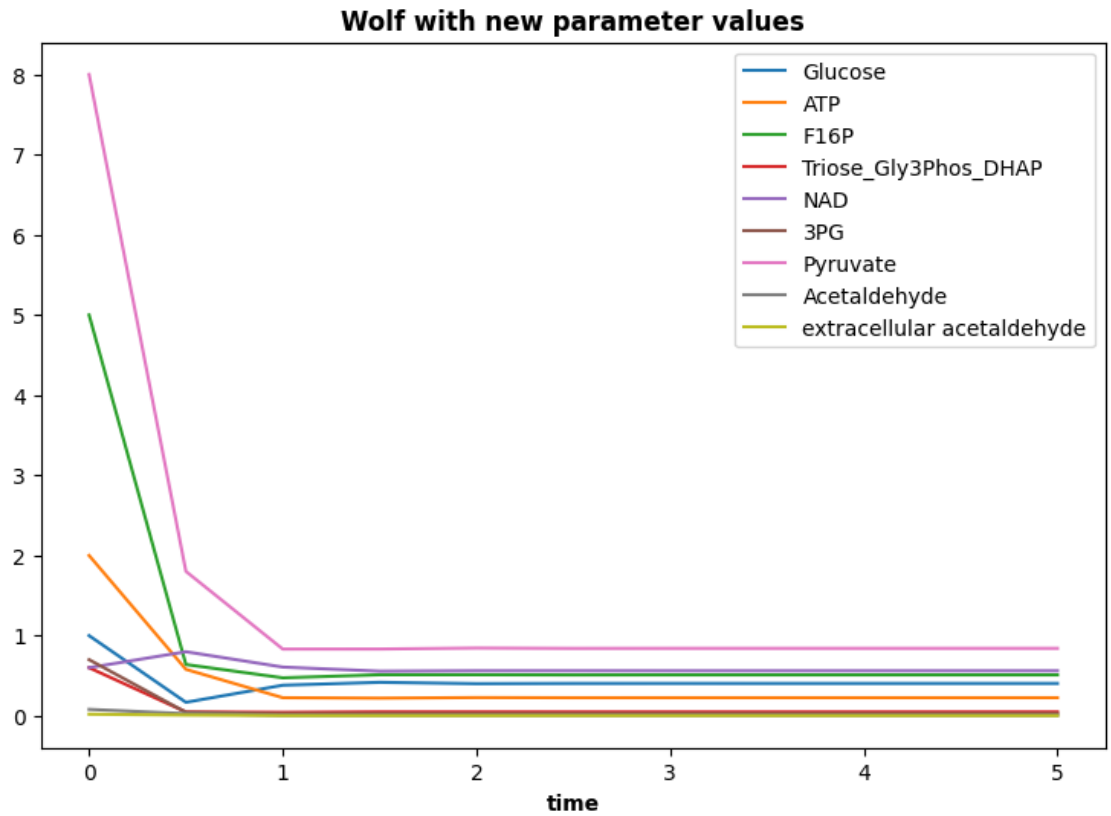
[13]:
```
smtc = ss.makeSingleModelTimeCourse(WOLF_URL, num_point=1000)
_ = smtc.execute()
```



[14]:
```
ss.getModelInformation(WOLF_URL)
```

[14]:
```
Model: Wolf2000_Glycolytic_Oscillations
Parameters: {'k0': 50.0, 'k1': 550.0, 'k2': 9.8, 'k31': 323.8, 'k33': 57823.1,
'k32': 76411.1, 'k34': 23.7, 'k4': 80.0, 'k5': 9.7, 'k6': 2000.0, 'k7': 28.0,
'k8': 85.7, 'k9': 80.0, 'k10': 375.0, 'atot': 4.0, 'ntot': 1.0, 'n': 4.0, 'ki':
1.0}
Floating Species: {'s1': 1.0, 'at': 2.0, 's2': 5.0, 's3': 0.6, 'na': 0.6, 's4':
0.7, 's5': 8.0, 's6': 0.08, 's6o': 0.02}
Boundary Species: {}
Number of Reactions: 11
Number of Species: 9
```

[15]:
```
smtc = ss.makeSingleModelTimeCourse(WOLF_URL, model_parameter_dct=dict(k0=5,␣
↪k1=55),
            title="Wolf with new parameter values")
df = smtc.execute()
```



[16]:
```
df.head()
```

[16]:
```
   time  project_task.s1  project_task.at  project_task.s2  project_task.s3  \
0   0.0         1.000000         2.000000         5.000000         0.600000
1   0.5         0.167970         0.579162         0.639675         0.047800
2   1.0         0.381272         0.225642         0.473636         0.042924
3   1.5         0.417602         0.220384         0.511794         0.049333
4   2.0         0.400847         0.227533         0.511717         0.049070

   project_task.na  project_task.s4  project_task.s5  project_task.s6  \
0         0.600000         0.700000         8.000000         0.080000
1         0.799205         0.043673         1.801447         0.026906
2         0.607620         0.025901         0.832152         0.007774
3         0.556435         0.026954         0.832856         0.007070
4         0.561399         0.027128         0.845115         0.007238
```

```
   project_task.s6o
0          0.020000
1          0.008861
2          0.002500
3          0.002255
4          0.002310
```

[17]:
```
smtc.cleanUp()
```

4 Use Case 2: Multiple Models Time Course

When comparing the time course simulation of multiple models, we want to compare selected **display variables** from the models, either in plots or in a DataFrame. This is done with the `MultipleModelTimeCourse` object.

[18]:
```
mmtc = ss.makeMultipleModelTimeCourse([MODEL_ANT, MODEL2_ANT],␣
↪model_parameter_dct=dict(k1=10), num_point=100)
```

[19]:
```
print(mmtc.getSEDML())
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Created by phraSED-ML version v1.3.0 with libSBML version 5.19.5. -->
<sedML xmlns="http://sed-ml.org/sed-ml/level1/version4"
xmlns:sbml="http://www.sbml.org/sbml/level3/version2/core" level="1"
version="4">
  <listOfModels>
    <model id="model0" language="urn:sedml:language:sbml.level-3.version-2" sour
ce="/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model0.xml">
      <listOfChanges>
        <changeAttribute target="/sbml:sbml/sbml:model/sbml:listOfParameters/sbm
l:parameter[@id='k1']/@value" newValue="10"/>
      </listOfChanges>
    </model>
    <model id="model1" language="urn:sedml:language:sbml.level-3.version-2" sour
ce="/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model1.xml">
      <listOfChanges>
        <changeAttribute target="/sbml:sbml/sbml:model/sbml:listOfParameters/sbm
l:parameter[@id='k1']/@value" newValue="10"/>
      </listOfChanges>
    </model>
  </listOfModels>
  <listOfSimulations>
    <uniformTimeCourse id="mmtc_sim1" initialTime="0" outputStartTime="0"
outputEndTime="5" numberOfSteps="99">
      <algorithm name="CVODE" kisaoID="KISAO:0000019"/>
    </uniformTimeCourse>
  </listOfSimulations>
```

```
  <listOfTasks>
    <task id="ta_model0" modelReference="model0"
simulationReference="mmtc_sim1"/>
    <task id="ta_model1" modelReference="model1"
simulationReference="mmtc_sim1"/>
  </listOfTasks>
  <listOfDataGenerators>
    <dataGenerator id="report_0_0_0" name="time">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> ta_model0_____time </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model0_____time" symbol="urn:sedml:symbol:time"
taskReference="ta_model0" modelReference="model0"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_1" name="model0">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> ta_model0_____S1 </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model0_____S1"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S1']"
taskReference="ta_model0" modelReference="model0"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_2" name="model1">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> ta_model1_____S1 </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model1_____S1"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S1']"
taskReference="ta_model1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_3" name="model0">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> ta_model0_____S2 </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model0_____S2"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S2']"
taskReference="ta_model0" modelReference="model0"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_4" name="model1">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
```

```
        <ci> ta_model1_____S2 </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model1_____S2"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S2']"
taskReference="ta_model1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_5" name="model0">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> ta_model0_____S3 </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model0_____S3"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S3']"
taskReference="ta_model0" modelReference="model0"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_6" name="model1">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> ta_model1_____S3 </ci>
      </math>
      <listOfVariables>
        <variable id="ta_model1_____S3"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S3']"
taskReference="ta_model1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
  </listOfDataGenerators>
  <listOfOutputs>
    <report id="report_0">
      <listOfDataSets>
        <dataSet id="report_0_0_0_dataset" label="ta_model0.time"
dataReference="report_0_0_0"/>
        <dataSet id="report_0_0_1_dataset" label="ta_model0.S1"
dataReference="report_0_0_1"/>
        <dataSet id="report_0_0_2_dataset" label="ta_model1.S1"
dataReference="report_0_0_2"/>
        <dataSet id="report_0_0_3_dataset" label="ta_model0.S2"
dataReference="report_0_0_3"/>
        <dataSet id="report_0_0_4_dataset" label="ta_model1.S2"
dataReference="report_0_0_4"/>
        <dataSet id="report_0_0_5_dataset" label="ta_model0.S3"
dataReference="report_0_0_5"/>
        <dataSet id="report_0_0_6_dataset" label="ta_model1.S3"
dataReference="report_0_0_6"/>
      </listOfDataSets>
    </report>
```

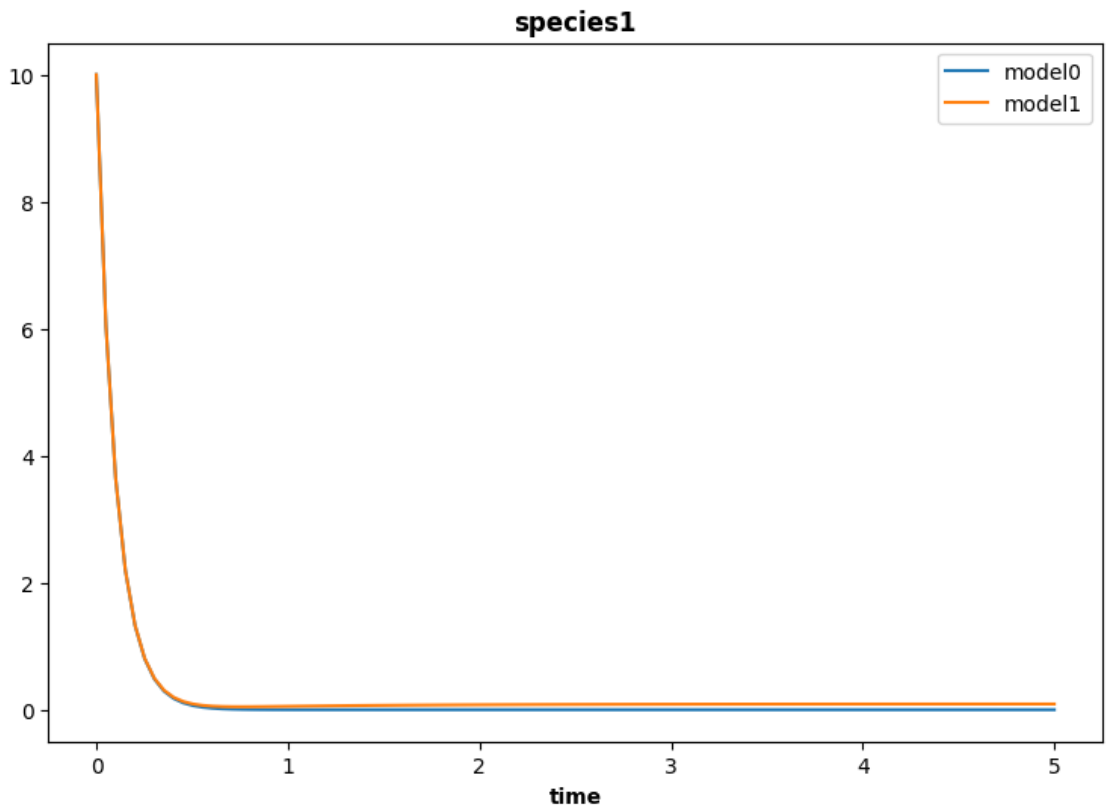
```
    <plot2D id="plot_1" name="species1">
      <listOfCurves>
        <curve id="plot_1__report_0_0_0__report_0_0_1" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_1"/>
        <curve id="plot_1__report_0_0_0__report_0_0_2" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_2"/>
      </listOfCurves>
    </plot2D>
    <plot2D id="plot_2" name="species2">
      <listOfCurves>
        <curve id="plot_2__report_0_0_0__report_0_0_3" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_3"/>
        <curve id="plot_2__report_0_0_0__report_0_0_4" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_4"/>
      </listOfCurves>
    </plot2D>
    <plot2D id="plot_3" name="S3">
      <listOfCurves>
        <curve id="plot_3__report_0_0_0__report_0_0_5" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_5"/>
        <curve id="plot_3__report_0_0_0__report_0_0_6" logX="false"
xDataReference="report_0_0_0" logY="false" yDataReference="report_0_0_6"/>
      </listOfCurves>
    </plot2D>
  </listOfOutputs>
</sedML>
```

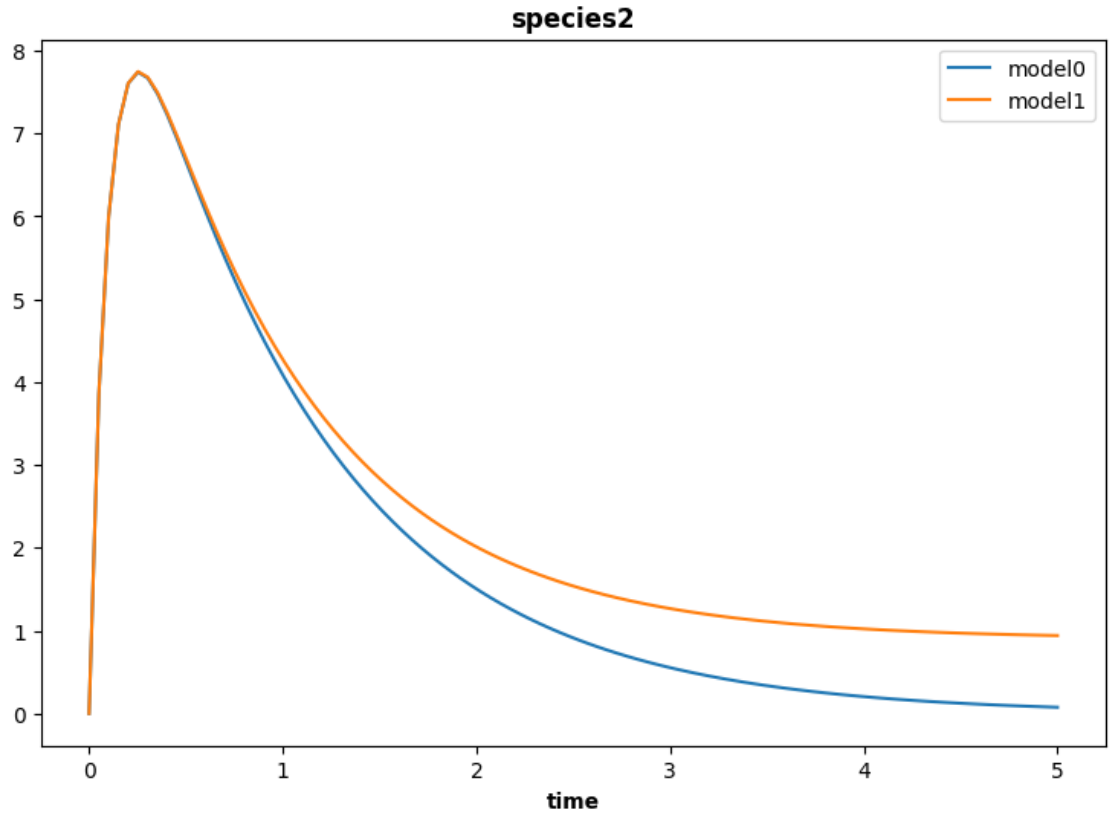
[20]:
```
print(mmtc.getPhraSEDML())
```

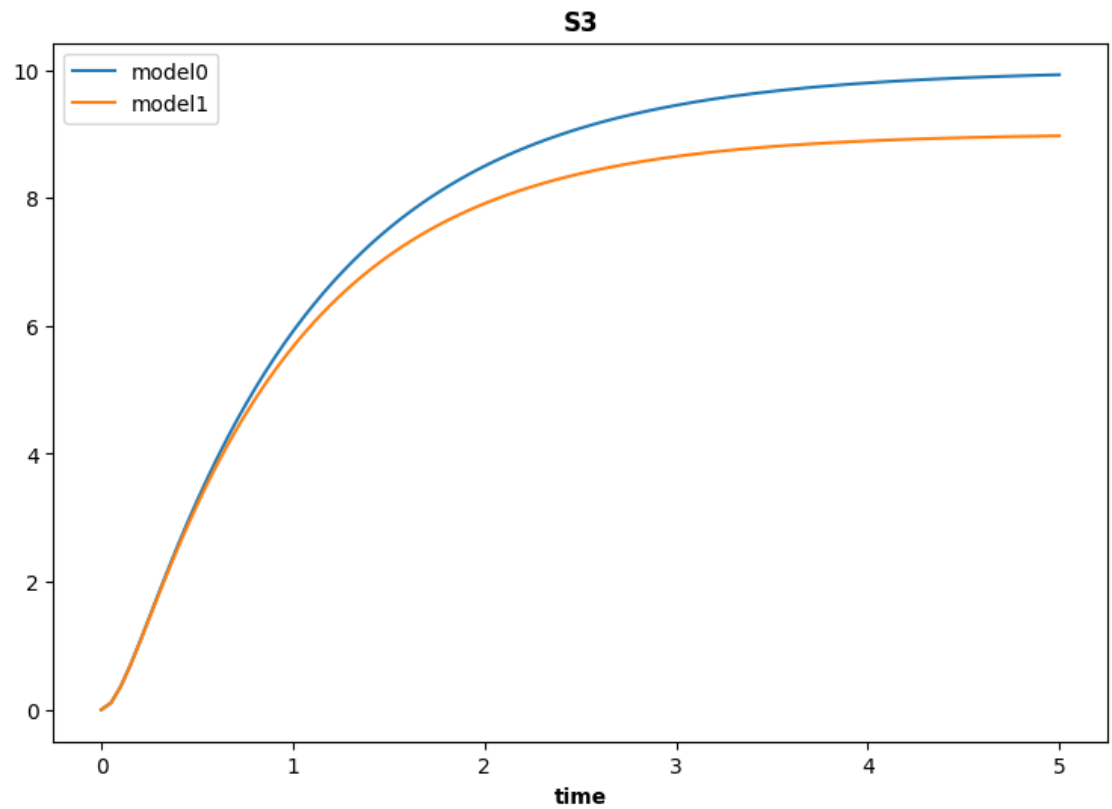
```
model0 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model0.xml"  with
k1 = 10
model1 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model1.xml"  with
k1 = 10
model2 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model2.xml"  with
k1 = 10
model3 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model3.xml"  with
k1 = 10
mmtc_sim1 = simulate uniform(0.0, 5.0, 99)
mmtc_sim1.algorithm = CVODE
ta_model0 = run mmtc_sim1 on model0
ta_model1 = run mmtc_sim1 on model1
ta_model2 = run mmtc_sim1 on model2
ta_model3 = run mmtc_sim1 on model3
```

```
report ta_model0.time, ta_model0.S1, ta_model1.S1, ta_model0.S2, ta_model1.S2,
ta_model0.S3, ta_model1.S3
plot "species1" ta_model0.time vs ta_model0.S1, ta_model1.S1
plot "species2" ta_model0.time vs ta_model0.S2, ta_model1.S2
plot "S3" ta_model0.time vs ta_model0.S3, ta_model1.S3
```

[21]:
```
mmtc.execute()
```




species2
model0
model1
8
7
6
5
4
3
2
1
0
0
1
2
3
4
5
time



[21]:

```
     ta_model0.time  ta_model0.S1  ta_model1.S1  ta_model0.S2  ta_model1.S2  \
0          0.000000  1.000000e+01     10.000000      0.000000      0.000000
1          0.050505  6.034751e+00      6.034912      3.858601      3.858627
2          0.101010  3.641824e+00      3.642859      5.997128      5.997406
3          0.151515  2.197752e+00      2.200499      7.106996      7.108161
4          0.202020  1.326293e+00      1.331475      7.604990      7.608033
..              …             …             …             …             …
95         4.797980 -6.086515e-14      0.089539      0.091627      0.950471
96         4.848485  2.622485e-14      0.089569      0.087114      0.947766
97         4.898990  5.909949e-14      0.089597      0.082824      0.945208
98         4.949495  7.401011e-14      0.089624      0.078745      0.942790
99         5.000000  4.876985e-14      0.089650      0.074866      0.940504

     ta_model0.S3  ta_model1.S3
0        0.000000      0.000000
1        0.106649      0.106461
2        0.361048      0.359735
3        0.695252      0.691340
4        1.068717      1.060492
..             …             …
95       9.908373      8.959990
```

```
96      9.912886      8.962665
97      9.917176      8.965194
98      9.921255      8.967585
99      9.925133      8.969846

[100 rows x 7 columns]
```

[22]:
```
mmtc.model_dct
```

[22]:
```
{'model0': <SimpleSEDML.model.Model at 0x11e080750>,
 'model1': <SimpleSEDML.model.Model at 0x11dfdbad0>,
 'model2': <SimpleSEDML.model.Model at 0x11e076050>,
 'model3': <SimpleSEDML.model.Model at 0x11dfa8890>,
 'model4': <SimpleSEDML.model.Model at 0x11e0438d0>,
 'model5': <SimpleSEDML.model.Model at 0x11e0d3f10>,
 'model6': <SimpleSEDML.model.Model at 0x11e036b50>,
 'model7': <SimpleSEDML.model.Model at 0x11e0a6d90>}
```

[23]:
```
print(mmtc.getPhraSEDML())
```

```
model0 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model0.xml"  with
k1 = 10
model1 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model1.xml"  with
k1 = 10
model2 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model2.xml"  with
k1 = 10
model3 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model3.xml"  with
k1 = 10
model4 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model4.xml"  with
k1 = 10
model5 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model5.xml"  with
k1 = 10
model6 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model6.xml"  with
k1 = 10
model7 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model7.xml"  with
k1 = 10
model8 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model8.xml"  with
k1 = 10
```

```
model9 = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmptpfccc8v/model9.xml"  with
k1 = 10
mmtc_sim1 = simulate uniform(0.0, 5.0, 99)
mmtc_sim1.algorithm = CVODE
ta_model0 = run mmtc_sim1 on model0
ta_model1 = run mmtc_sim1 on model1
ta_model2 = run mmtc_sim1 on model2
ta_model3 = run mmtc_sim1 on model3
ta_model4 = run mmtc_sim1 on model4
ta_model5 = run mmtc_sim1 on model5
ta_model6 = run mmtc_sim1 on model6
ta_model7 = run mmtc_sim1 on model7
ta_model8 = run mmtc_sim1 on model8
ta_model9 = run mmtc_sim1 on model9
report ta_model0.time, ta_model0.S1, ta_model1.S1, ta_model0.S2, ta_model1.S2,
ta_model0.S3, ta_model1.S3
plot "species1" ta_model0.time vs ta_model0.S1, ta_model1.S1
plot "species2" ta_model0.time vs ta_model0.S2, ta_model1.S2
plot "S3" ta_model0.time vs ta_model0.S3, ta_model1.S3
```

4.1 Creating an OMEX file

We can create an OMEX file for this experiment.

[24]:
```
omex_path, _ = mmtc.makeOMEXFile()
omex_path
```

[24]: '/Users/jlheller/home/Technical/repos/SimpleSEDML/examples/project.omex'

[25]:
```
# We can examine the content of the OMEX file.
!unzip -l project.omex
```

```
Archive:  project.omex
  Length      Date    Time    Name
---------  ---------- -----   ----
     1163  04-11-2026 09:01   metadata.rdf
     2261  04-11-2026 09:01   model0.xml
     2844  04-11-2026 09:01   model1.xml
     2261  04-11-2026 09:01   model10.xml
     2844  04-11-2026 09:01   model11.xml
     2261  04-11-2026 09:01   model2.xml
     2844  04-11-2026 09:01   model3.xml
     2261  04-11-2026 09:01   model4.xml
     2844  04-11-2026 09:01   model5.xml
     2261  04-11-2026 09:01   model6.xml
     2844  04-11-2026 09:01   model7.xml
     2261  04-11-2026 09:01   model8.xml
```

```
     2844  04-11-2026 09:01   model9.xml
    10375  04-11-2026 09:01   project.sedml
     1695  04-11-2026 09:01   manifest.xml
---------                     -------
    43863                     15 files
```

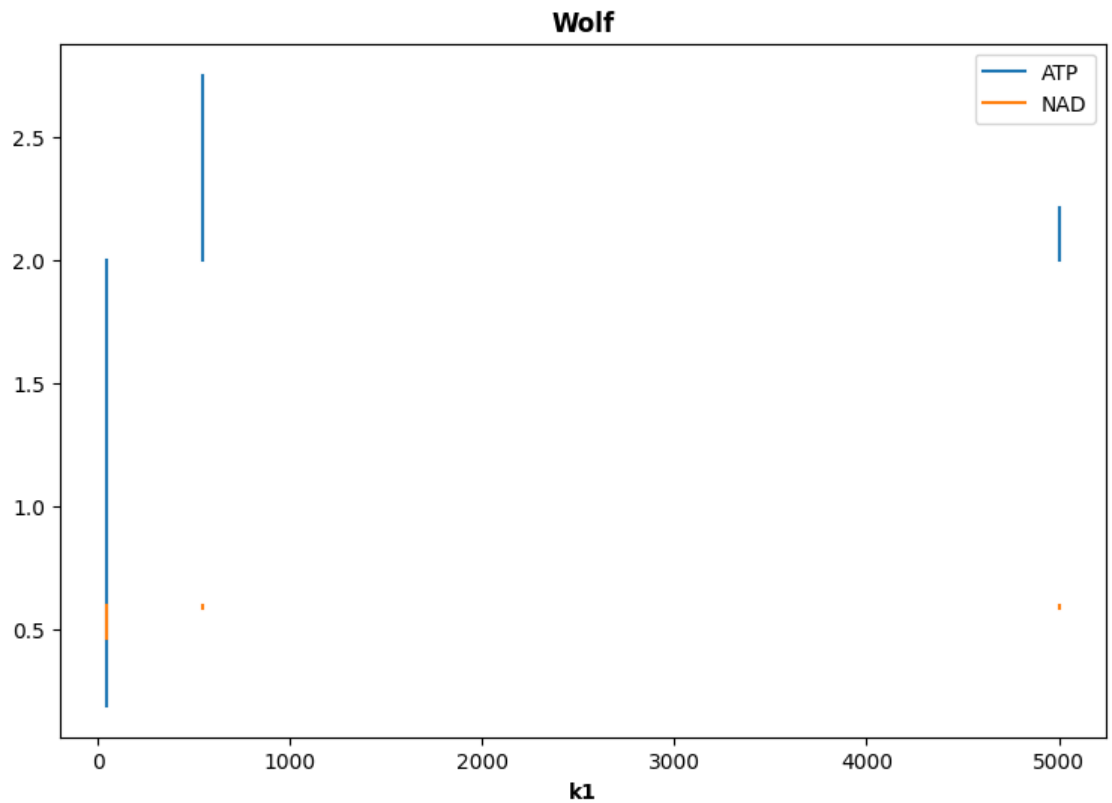
[26]:
```
mmtc.cleanUp()
```

5 Use Case 3: Single Model Parameter Scan

[27]:
```
smps = ss.makeSingleModelParameterScan(WOLF_URL, simulation_type="onestep",
                                       project_id="Wolf", title="Wolf",
        time_interval=10, display_variables=["at", "na"],␣
 ↪scan_parameter_dct=dict(k1=[50, 550, 5000]))
print(smps.getPhraSEDML())
_ = smps.execute()
```

```
md_Wolf = model
"/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmppkl583lq/md_Wolf.xml"
si_Wolf = simulate onestep(10)
st_Wolf = run si_Wolf on md_Wolf
repeated_taskWolf = repeat st_Wolf for k1 in [50, 550, 5000], reset=true
report repeated_taskWolf.k1, repeated_taskWolf.at, repeated_taskWolf.na
plot "Wolf" repeated_taskWolf.k1 vs repeated_taskWolf.at, repeated_taskWolf.na
```

```
/Users/jlheller/home/Technical/repos/SimpleSEDML/src/SimpleSEDML/simple_sedml.py
:398: UserWarning: Reports only generate data for the last repeated task.
  warnings.warn("Reports only generate data for the last repeated task.")
```



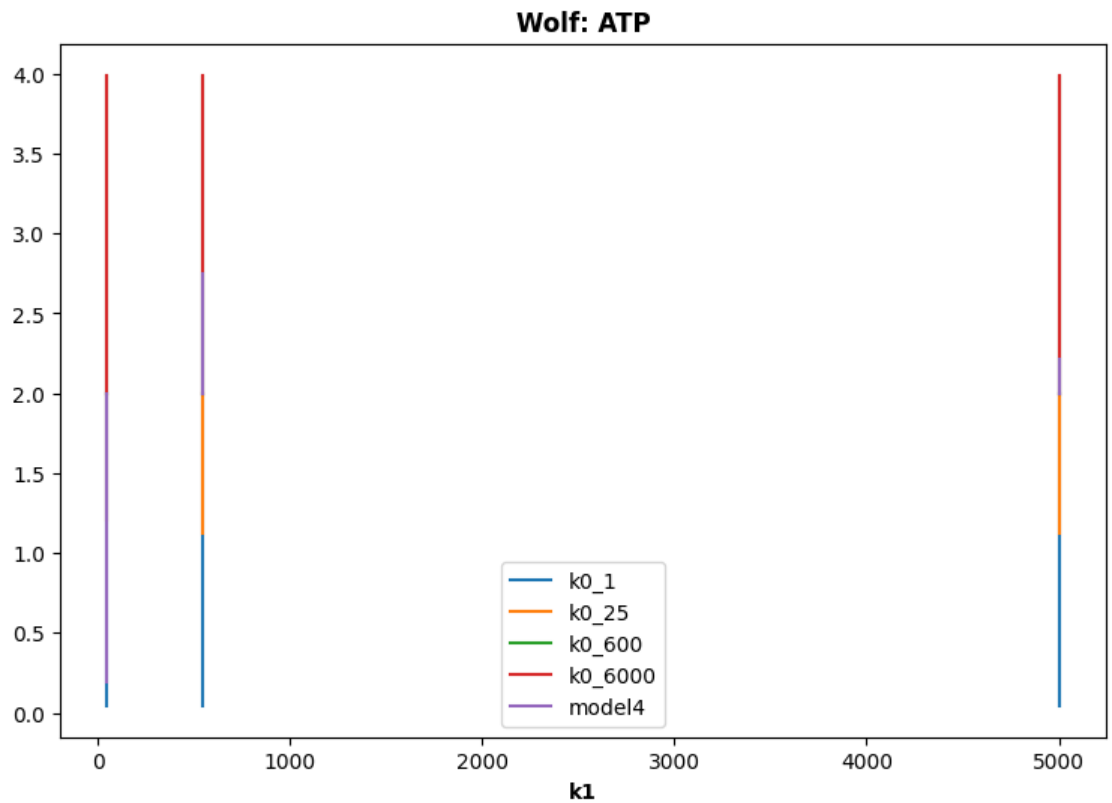
6 Use Case 4: Multiple Model Parameter Scan

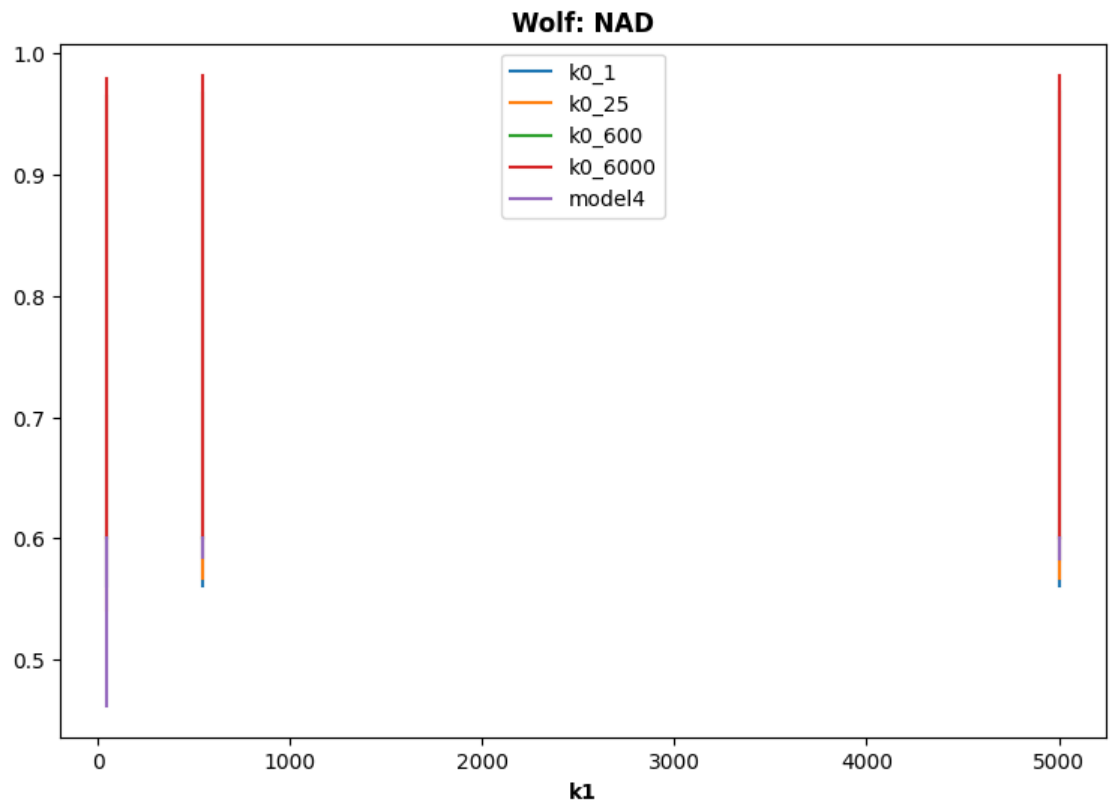
[28]:
```
scan_parameter_df = pd.DataFrame(dict(k1=[50, 550, 5000]))
mmps = ss.makeMultipleModelParameterScan(scan_parameter_df,
                                         model_refs=[WOLF_URL],
                                         simulation_type="onestep",
                                         project_id="Wolf",
                                         title="Wolf",
                                         time_interval=10,
                                         display_variables=['at', 'na'])
```

[29]:
```
# We create additional models and they will be included in the simulations,␣
↪plots and reports
_ = mmps.addModel("k0_1", WOLF_URL, parameter_dct=dict(k0=1))
_ = mmps.addModel("k0_25", WOLF_URL, parameter_dct=dict(k0=25))
_ = mmps.addModel("k0_600", WOLF_URL, parameter_dct=dict(k0=600))
_ = mmps.addModel("k0_6000", WOLF_URL, parameter_dct=dict(k0=6000))
```

[30]:
```
#print(smps.getPhraSEDML())
_ = mmps.execute()
```

```
/Users/jlheller/home/Technical/repos/SimpleSEDML/src/SimpleSEDML/multiple_model_
simple_sedml.py:212: UserWarning: Can only plot one parameter on the x-axis.
  warnings.warn("Can only plot one parameter on the x-axis. ")
```





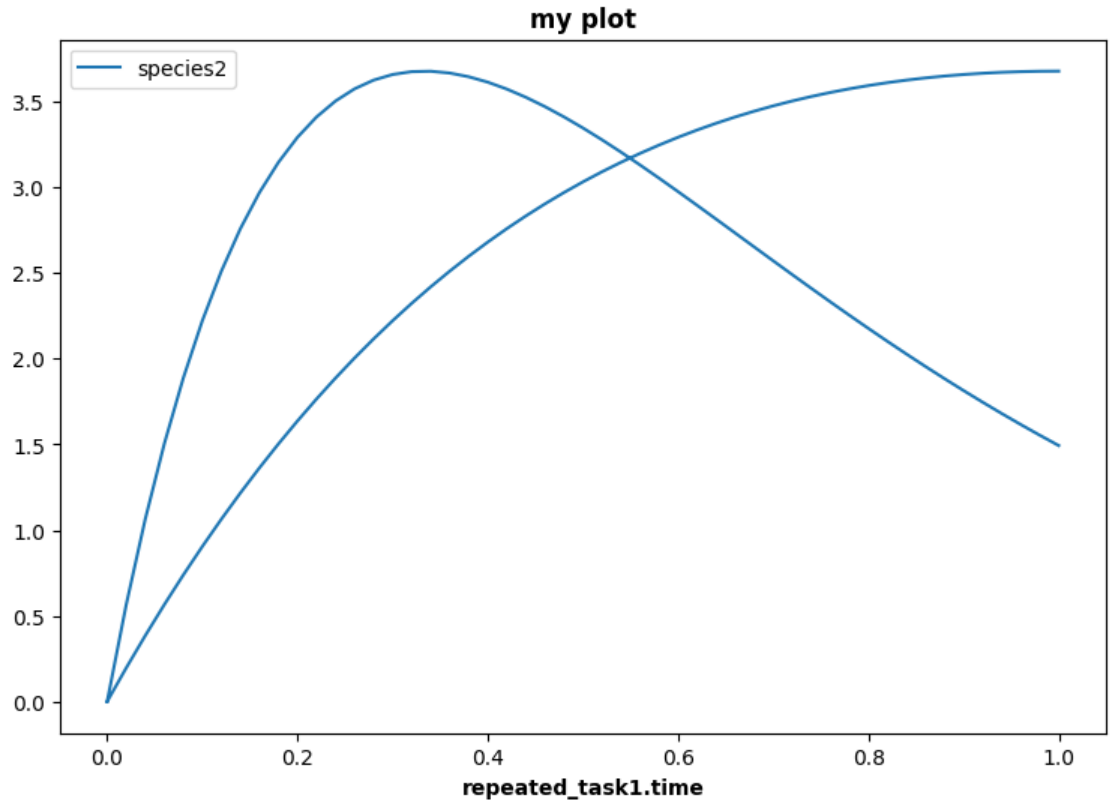
7 Low Level API

Below, we illustrate use of the low level API in isolation.

[31]:
```
# Make DataFrame of changes to model parameters
parameter_df = pd.DataFrame(dict(k1=[1, 3], k2=[1, 3]))
```

[32]:
```
ssed = ss.SimpleSEDML()
ssed.addModel("model1", MODEL_ANT, parameter_dct=dict(k1=3),␣
 ↪ref_type="ant_str", is_overwrite=True)
ssed.addSimulation("sim1", "uniform", start=0, end=1, num_step=50)
ssed.addTask("task1", "model1", "sim1")
ssed.addRepeatedTask("repeated_task1", "task1", parameter_df=parameter_df)
ssed.addReport("task1.time", "task1.S1", "task1.S2", "task1.S3")
ssed.addPlot("repeated_task1.time", "repeated_task1.S2", title="my plot")
result_df = ssed.execute()
```

```
/Users/jlheller/home/Technical/repos/SimpleSEDML/src/SimpleSEDML/simple_sedml.py
:398: UserWarning: Reports only generate data for the last repeated task.
  warnings.warn("Reports only generate data for the last repeated task.")
```



[33]:
```
print(ssed.getSEDML())
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Created by phraSED-ML version v1.3.0 with libSBML version 5.19.5. -->
<sedML xmlns="http://sed-ml.org/sed-ml/level1/version4"
xmlns:sbml="http://www.sbml.org/sbml/level3/version2/core" level="1"
version="4">
  <listOfModels>
    <model id="model1" language="urn:sedml:language:sbml.level-3.version-2" sour
ce="/var/folders/02/lb248q1j50dch2pthzkvxmpr0000gn/T/tmpu7a6g_dg/model1.xml">
      <listOfChanges>
        <changeAttribute target="/sbml:sbml/sbml:model/sbml:listOfParameters/sbm
l:parameter[@id='k1']/@value" newValue="3"/>
      </listOfChanges>
    </model>
  </listOfModels>
  <listOfSimulations>
    <uniformTimeCourse id="sim1" initialTime="0" outputStartTime="0"
outputEndTime="1" numberOfSteps="50">
      <algorithm name="CVODE" kisaoID="KISAO:0000019"/>
    </uniformTimeCourse>
  </listOfSimulations>
```

```
  <listOfTasks>
    <task id="task1" modelReference="model1" simulationReference="sim1"/>
    <repeatedTask id="repeated_task1" range="vector_for_k2" resetModel="true">
      <listOfRanges>
        <vectorRange id="vector_for_k1">
          <value> 1 </value>
          <value> 3 </value>
        </vectorRange>
        <vectorRange id="vector_for_k2">
          <value> 1 </value>
          <value> 3 </value>
        </vectorRange>
      </listOfRanges>
      <listOfChanges>
        <setValue modelReference="model1"
target="/sbml:sbml/sbml:model/sbml:listOfParameters/sbml:parameter[@id='k1']"
range="vector_for_k1">
          <math xmlns="http://www.w3.org/1998/Math/MathML">
            <ci> vector_for_k1 </ci>
          </math>
        </setValue>
        <setValue modelReference="model1"
target="/sbml:sbml/sbml:model/sbml:listOfParameters/sbml:parameter[@id='k2']"
range="vector_for_k2">
          <math xmlns="http://www.w3.org/1998/Math/MathML">
            <ci> vector_for_k2 </ci>
          </math>
        </setValue>
      </listOfChanges>
      <listOfSubTasks>
        <subTask order="0" task="task1"/>
      </listOfSubTasks>
    </repeatedTask>
  </listOfTasks>
  <listOfDataGenerators>
    <dataGenerator id="report_0_0_0" name="task1.time">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> task1_____time </ci>
      </math>
      <listOfVariables>
        <variable id="task1_____time" symbol="urn:sedml:symbol:time"
taskReference="task1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_1" name="species1">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> task1_____S1 </ci>
      </math>
```

```
      <listOfVariables>
        <variable id="task1_____S1"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S1']"
taskReference="task1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_2" name="species2">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> task1_____S2 </ci>
      </math>
      <listOfVariables>
        <variable id="task1_____S2"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S2']"
taskReference="task1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="report_0_0_3" name="S3">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> task1_____S3 </ci>
      </math>
      <listOfVariables>
        <variable id="task1_____S3"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S3']"
taskReference="task1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="plot_1_0_0" name="repeated_task1.time">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> repeated_task1_____time </ci>
      </math>
      <listOfVariables>
        <variable id="repeated_task1_____time" symbol="urn:sedml:symbol:time"
taskReference="repeated_task1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
    <dataGenerator id="plot_1_0_1" name="species2">
      <math xmlns="http://www.w3.org/1998/Math/MathML">
        <ci> repeated_task1_____S2 </ci>
      </math>
      <listOfVariables>
        <variable id="repeated_task1_____S2"
target="/sbml:sbml/sbml:model/sbml:listOfSpecies/sbml:species[@id='S2']"
taskReference="repeated_task1" modelReference="model1"/>
      </listOfVariables>
    </dataGenerator>
  </listOfDataGenerators>
  <listOfOutputs>
    <report id="report_0">
```

```
      <listOfDataSets>
        <dataSet id="report_0_0_0_dataset" label="task1.time"
dataReference="report_0_0_0"/>
        <dataSet id="report_0_0_1_dataset" label="task1.S1"
dataReference="report_0_0_1"/>
        <dataSet id="report_0_0_2_dataset" label="task1.S2"
dataReference="report_0_0_2"/>
        <dataSet id="report_0_0_3_dataset" label="task1.S3"
dataReference="report_0_0_3"/>
      </listOfDataSets>
    </report>
    <plot2D id="plot_1" name="my plot">
      <listOfCurves>
        <curve id="plot_1__plot_1_0_0__plot_1_0_1" logX="false"
xDataReference="plot_1_0_0" logY="false" yDataReference="plot_1_0_1"/>
      </listOfCurves>
    </plot2D>
  </listOfOutputs>
</sedML>
```

[34]:
```
ssed.cleanUp()
```

7.1 Special Executor

A special executor is provied to execute the SimpleSEDML directives without generating SED-ML. The executor's only argument is an SimpleSEDML object.

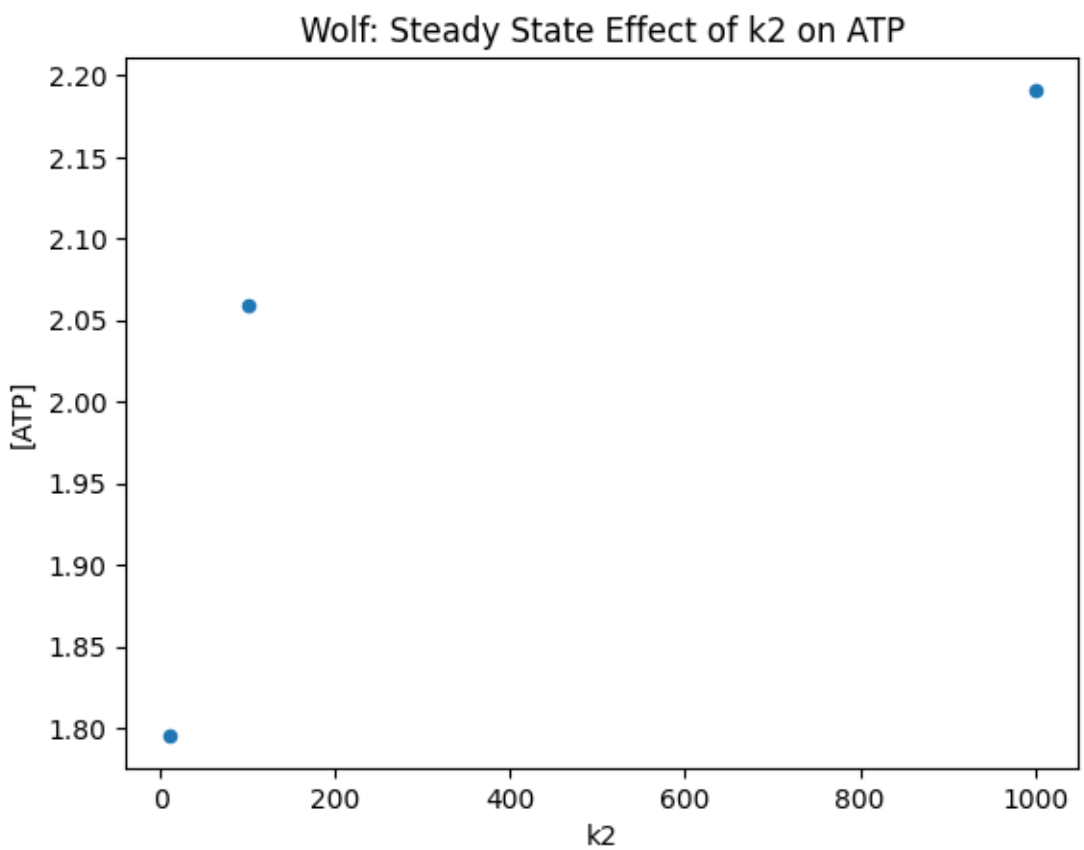
[35]:
```
ss.getModelInformation(ss.constants.WOLF_URL)
```

[35]:
```
Model: Wolf2000_Glycolytic_Oscillations
Parameters: {'k0': 50.0, 'k1': 550.0, 'k2': 9.8, 'k31': 323.8, 'k33': 57823.1,
'k32': 76411.1, 'k34': 23.7, 'k4': 80.0, 'k5': 9.7, 'k6': 2000.0, 'k7': 28.0,
'k8': 85.7, 'k9': 80.0, 'k10': 375.0, 'atot': 4.0, 'ntot': 1.0, 'n': 4.0, 'ki':
1.0}
Floating Species: {'s1': 1.0, 'at': 2.0, 's2': 5.0, 's3': 0.6, 'na': 0.6, 's4':
0.7, 's5': 8.0, 's6': 0.08, 's6o': 0.02}
Boundary Species: {}
Number of Reactions: 11
Number of Species: 9
```

[36]:
```
scan_parameter_df = pd.DataFrame(dict(k2=[10, 100, 1000], k31=[100, 200, 300]))
mmps = ss.makeMultipleModelParameterScan(scan_parameter_df,
                                         model_refs=[WOLF_URL],
                                         simulation_type="steadystate",
                                         project_id="Wolf",
                                         title="Wolf",
                                         time_interval=10,
```

```
                                                display_variables=['at', 'na'])
executor = ss.Executor(mmps)
result = executor.executePlot2d(kind="scatter")
result.ax.set_title("Wolf: Steady State Effect of k2 on ATP")
executor.cleanUp()
```

```
/Users/jlheller/home/Technical/repos/SimpleSEDML/src/SimpleSEDML/multiple_model_
simple_sedml.py:212: UserWarning: Can only plot one parameter on the x-axis.
  warnings.warn("Can only plot one parameter on the x-axis. ")
/Users/jlheller/home/Technical/repos/SimpleSEDML/src/SimpleSEDML/executor.py:177
: UserWarning: Multiple plots available. Plotting the first.
  warnings.warn("Multiple plots available. Plotting the first.")
```



[37]:
```
result = executor.executeRepeatedTask()
result
```

[37]:

	at	na	k2	k31
0	1.795389	0.741899	10	100
1	2.058567	0.652585	100	200
2	2.191108	0.594709	1000	300

8 Cleanup

[38]:
```
import os
EXAMPLE_REMOVE_FILES = ["project.omex"]
example_dir = os.getcwd()
remove_files = [os.path.join(example_dir, f) for f in EXAMPLE_REMOVE_FILES]
for ffile in remove_files:
    if os.path.isfile(ffile):
        os.remove(ffile)
```